Machine Learning Principles

Class 20: November 13
Gaussian Mixture Modeling and EM algorithm

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Today's Lecture

1. **Learning** a Bayesian Net (learning parameters : CPT)
   - a network structure is known and data is fully observed
   - a network structure is known and data is partially observed

2. **E**xpectation **M**aximization (EM) algorithm (partial observation)

3. **G**aussian **M**ixture **M**odeling (GMM)
   - DAG structure & Application
   - learning GMM using EM

Learning a Bayesian Net

- we will focus on the case where a structure is given.

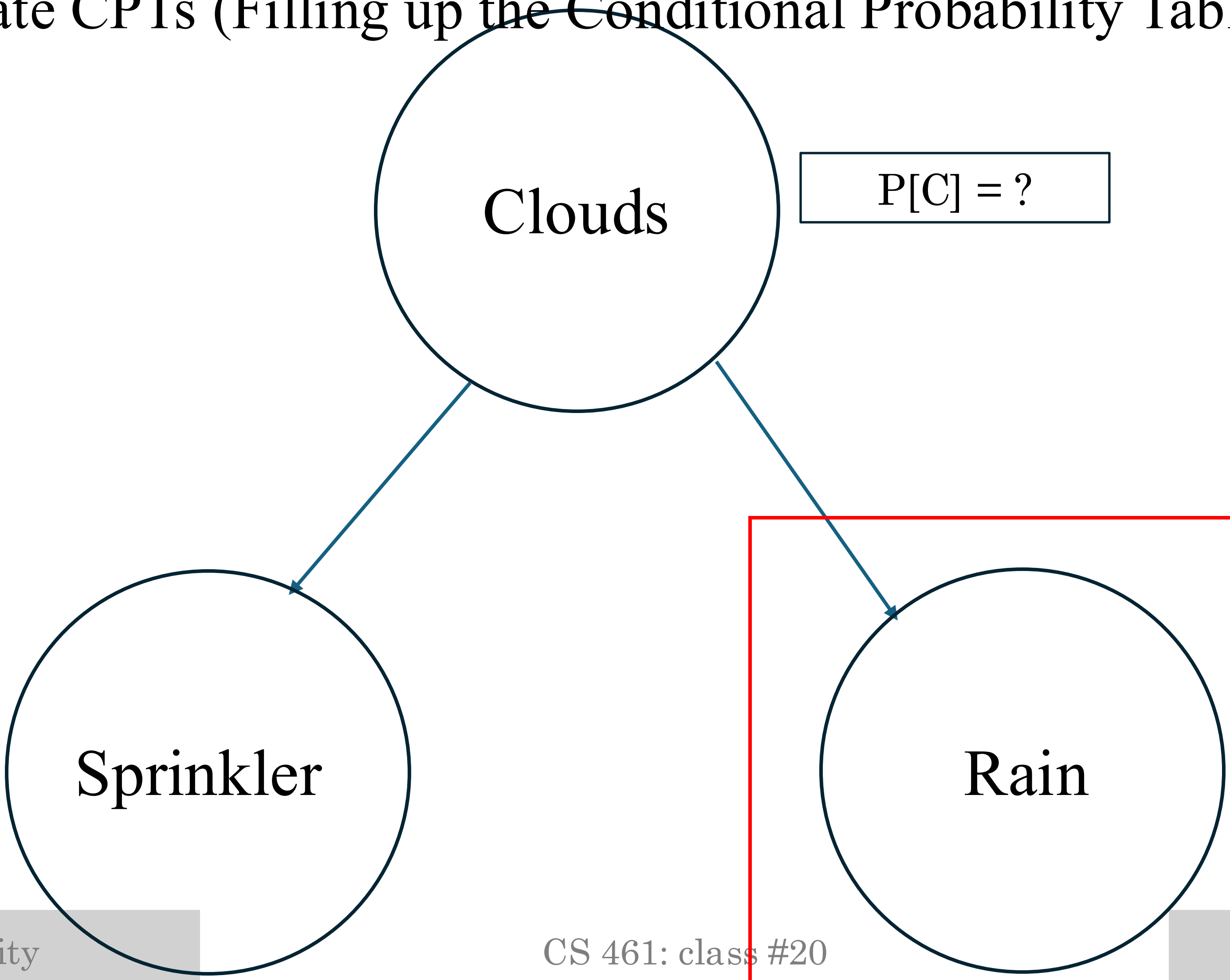
[1] Learning Bayesian Tree

If you are interested in learning a structure,
Chow-Liu Algorithm finds an optimal T (first / second order dependence)
T = $argmin_T \; KL\,(P||\,T)$, where P is a true distribution.

Q: what is the true distribution? How can we compute this?

[2] Learning Bayesian Tree (complete data observation)

When a structure is known and data is fully observed,
we need to estimate CPTs (Filling up the Conditional Probability Table).



C	P[S | C]
T	?
F	?

C	P[R | C]
T	? p
F	? q

[3] Learning Bayesian Tree (complete data observation)

MLE estimation for p: P[R + | C +]
when data is fully observed.

$$p(D) = \prod_{n=1}^{N} P(x_n)$$

$$\log P(D) = \sum_{n=1}^{N} \log P(x_n)$$

$$= \sum_{n=1}^{N} \log P[C(x_n)] + \log P[S(x_n)|C(x_n)] + \boxed{\log P[R(x_n)|C(x_n)]}$$

the four possible cases of R and C
: $p\ 1-p, q, 1-q$

$$\frac{\partial \log P(D)}{\partial p} = N_p \cdot 1/P - N_n \cdot 1/(1-P) = 0$$

$$\boxed{P = \frac{N_p}{N_p + N_n}}$$

- N_P: # samples cloud + & rain +
- N_n: # samples cloud + & rain −

This is just a sample mean (Bernoulli R.V)

[4] Learning Bayesian Tree (partial data observation)

what if data is partially observed?
suppose we forgot to collect **cloud** information.
can we still estimate p *or* q : P[rain| cloud +] or P[rain| cloud -]?

[5] Learning Bayesian Tree (partial data observation)

- MLE when data is partially observed.

$$\log P(D|\theta,p,q) = \sum_{n=1}^{N} \log \sum_{C_n} P(S_n, R_n, C_n|\theta,p,q)$$

marginalization over unobserved variables

- MLE when data is fully observed (complete MLE)

$$\log P(D|\theta,p,q) = \sum_{n=1}^{N} \log P(S_n, R_n, C_n|\theta,p,q)$$

Q: which one is easier in computing MLE ?

[6] Learning Bayesian Tree (partial data observation)

$$\log P(D|\theta, p, q)) = \sum_{n=1}^{N} \log P(S_n, R_n, C_n|\theta, p, q)$$
$$= \sum_{n=1}^{N} \log P(C_n) + \log P(S_n|C_n) + \log P(R_n|C_n)$$

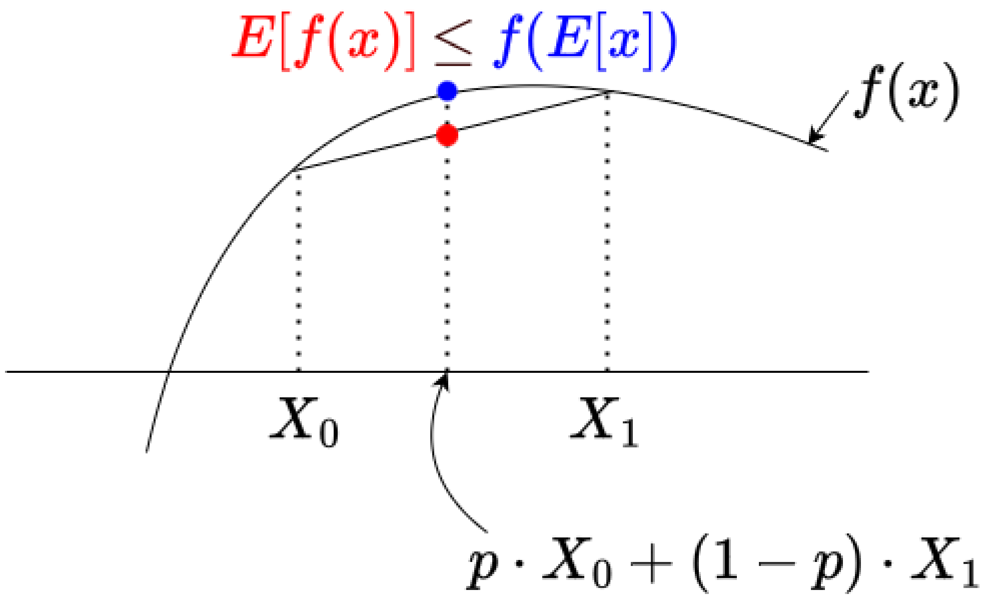
$$\log P(D|\theta, p, q) = \sum_{n=1}^{N} \log \sum_{C_n} P(S_n, R_n, C_n|\theta, p, q)$$
$$\boxed{= \sum_{n=1}^{N} \log(P(C_n+) \cdot P(S_n|C_n+) \cdot P(R_n|C_n+) + P(C_n-) \cdot P(S_n|C_n-) \cdot P(R_n|C_n-))}$$

hard to optimize (parameters are interdependent)!

[7] Learning Bayesian Tree (partial data observation)

It’s hard to get a closed form to compute MLE,
when there are missing variables.
(log \sum is not tractable for MLE; can we change it to \sum log like the complete case?)

[8] Learning Bayesian Tree (partial data observation)



Jensen's Inequality: (both expectation are finite)
E[log X] ≤ log E[X]

[9] Learning Bayesian Tree (partial data observation)

$$\log P(D|\theta) = \sum_{n=1}^{N} \log \sum_{C_n} P(S_n, R_n, C_n|\theta)$$

$$= \sum_{n=1}^{N} \log \sum_{C_n} \frac{P(S_n, R_n, C_n|\theta) q(C_n)}{q(C_n)}$$

$$= \sum_{n=1}^{N} \log E[\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}]$$

$$\geq \sum_{n=1}^{N} E[\log(\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)})]$$

$$\geq \sum_{n=1}^{N} E[\log P(S_n, R_n, C_n|\theta)] + H(q(C_n))$$

- hard to optimize!
- by Jensen's Inequality
- convex,
- easy to optimize!

expectation over

$q(C_n)$: *any arbitrary density for* C_n

gives a lower bound.

Q: how can we make the lower bound tight?

[10] Learning Bayesian Tree (partial data observation)

Q: What does the expectation of likelihood mean?
Likelihood is not a probability. It is a function of the parameters.
In the example below, the likelihood with one point data x_i is a function of $L(\mu, \sigma)$.

$$\mathcal{N}(x_i|\mu,\sigma^2) = \frac{1}{\sqrt{(2\pi\sigma^2)}} \exp\{\frac{-1}{2\sigma^2}(x_i-\mu)^2\}$$

What <u>if x_i is unobserved</u> and instead we know the density <u>x_i</u> $\sim$ p<u>(x_i)?</u>
we can measure the expectation treating μ *and* σ as constants.

$$E[f(X_i|\mu,\sigma)] = \sum_{x_i} f(x_i|\mu,\sigma)p(x_i)$$

[11] Learning Bayesian Tree (partial data observation)

Q: what density makes the lower bound tight?

$$\sum_{n=1}^{N} E[\log P(S_n, R_n, C_n|\theta] + H(q(C_n))$$

$$= \sum_{n=1}^{N} \sum_{C_n} q(C_n) \log(\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)})$$

$$= \sum_{n=1}^{N} \sum_{C_n} q(C_n) \log(\frac{P(C_n|S_n, R_n, \theta) P(S_n, R_n|\theta)}{q(C_n)})$$

$$= \sum_{n=1}^{N} (\sum_{C_n} q(C_n) \log \frac{P(C_n|S_n, R_n, \theta)}{q(C_n)} + \sum_{C_n} q(C_n) \log P(S_n, R_n|\theta))$$

$$\boxed{= \sum_{n=1}^{N} -KL(q(C_n)||P(C_n|S_n, R_n, \theta))} + \log P(S_n, R_n|\theta)$$

when $q(C_i) = P(C_i | S_i, R_i, \theta, p, q)$

the lower bound becomes equal to $\log P(S_i, R_i | \theta, p, q)$

[12] Learning Bayesian Tree (partial data observation)

$$
\begin{aligned}
\log P(D|\theta) &= \sum_{n=1}^{N} \log \sum_{C_n} P(S_n, R_n, C_n|\theta) \\
&= \sum_{n=1}^{N} \log \sum_{C_n} \frac{P(S_n, R_n, C_n|\theta) q(C_n)}{q(C_n)} \\
&= \sum_{n=1}^{N} \log E[\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)}] \\
&\geq \sum_{n=1}^{N} E[\log(\frac{P(S_n, R_n, C_n|\theta)}{q(C_n)})] \\
&\geq \sum_{n=1}^{N} E[\log P(S_n, R_n, C_n|\theta)] + H(q(C_n))
\end{aligned}
$$

expectation over $q(C_i) = P(C_i \,| S_i, R_i, \theta, p, q)$

what is the next ?

we do maximization over θ, p, q. why?

Expectation Maximization:
finding maximum likelihood when data is partially observed.

[1] EM algorithm (cloud, sprinkler, rain example)

1. start with arbitrary parameters: θ, p, q
2. compute $P(C_n \,|S_n, R_n, \theta, p, q)$ for $\forall\ n$
3. **E step**: compute

$$\sum_{n=1}^{N} E[\log P(S_n, R_n, C_n|\theta, p, q]$$

4. **M step**: Update $\theta, p, q = \ argmax_{\theta,p,q} \quad \sum_{n=1}^{N} E[\log P(S_n, R_n, C_n|\theta, p, q]$

5. go back to step 2.

[2] EM algorithm (cloud, sprinkler, rain example)

1. start with arbitrary parameters: θ, p, q
2. compute $P(C_i \mid S_i, R_i, \theta, p, q)$ for $\forall\ i$
3. E step: compute $\sum_{n=1}^{N} E[\log P(S_i, R_i, C_i|\theta, p, q)] = \mathrm{Q}\ (\theta, \theta^t)$

   Auxiliary Function

4. S step: Update $\theta, p, q = argmax_{\theta,p,q} \sum_{n=1}^{N} E[\log P(S_i, R_i, C_i|\theta, p, q)]$

5. go back to step 2.

$$\text{Auxiliary Function } Q(\theta, \theta^t) = \sum_{n=1}^{N} E[\log P(S_i, R_i, \underbrace{C_i | \theta, p, q}_{\theta})]$$

expectation $P(C_i | S_i, R_i, \theta^t)$

- $Q(\theta^t, \theta^t) = \sum_i \log P[S_i, R_i | \theta^t]$
- $Q(\theta^{t+1}, \theta^t) \geq Q(\theta^t, \theta^t)$ by **Maximization**

EM monotonically increases the observed data log likelihood!

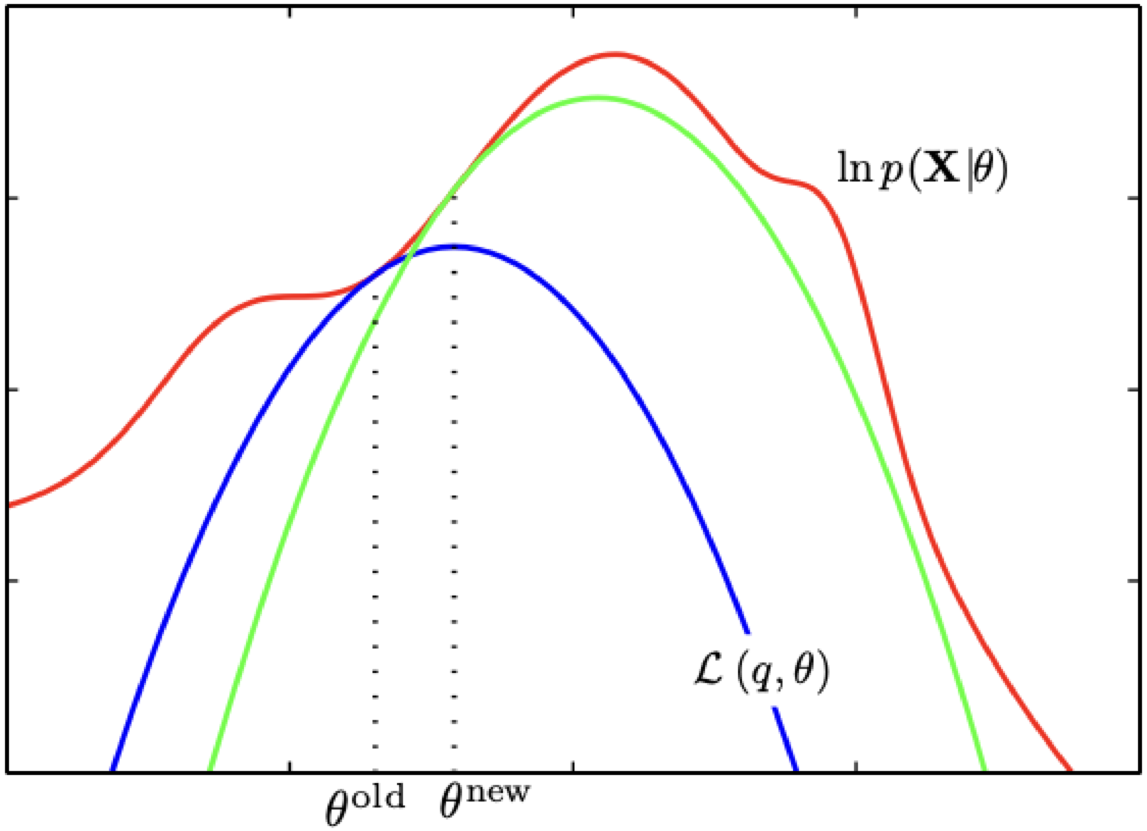
- $Q(\theta^{t+1}, \theta^{t+1}) \geq Q(\theta^{t+1}, \theta^t)$ by **lower bound**
- $Q(\theta^{t+1}, \theta^{t+1}) = \sum_i \log P[S_i, R_i | \theta^{t+1}]$ by **Expectation**
- $\sum_i \log P[S_i, R_i | \theta^t] \leq (\theta^{t+1}, \theta^t) \leq Q(\theta^{t+1}, \theta^{t+1}) = \sum_i \log P[S_i, R_i | \theta^{t+1}]$

[3] EM algorithm (original likelihood vs. auxiliary function)

EM algorithm finds a local minimum.
$\log P(X|\theta)$ is often non − convex or non − concave.

Figure 9.14 The EM algorithm involves alternately computing a lower bound on the log likelihood for the current parameter values and then maximizing this bound to obtain the new parameter values. See the text for a full discussion.

From textbook Bishop

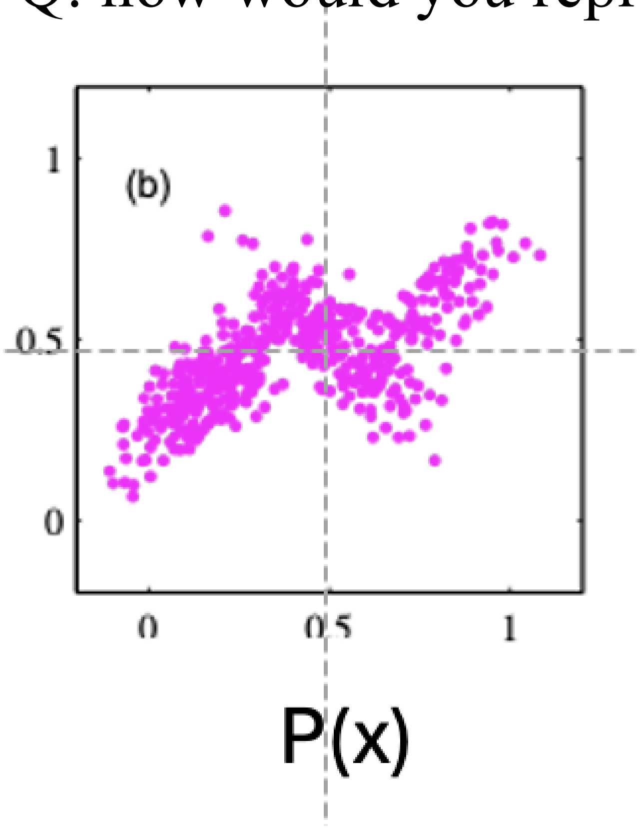


- Comparison between EM and gradient based methods

Gaussian Mixture Model (GMM)

[1] GMM (motivation)

Data often has multiple modalities like the figure below.
Q: continuous / discrete? How many dimensions?
Q: how would you represent the density $f(x_1, x_2)$: ?

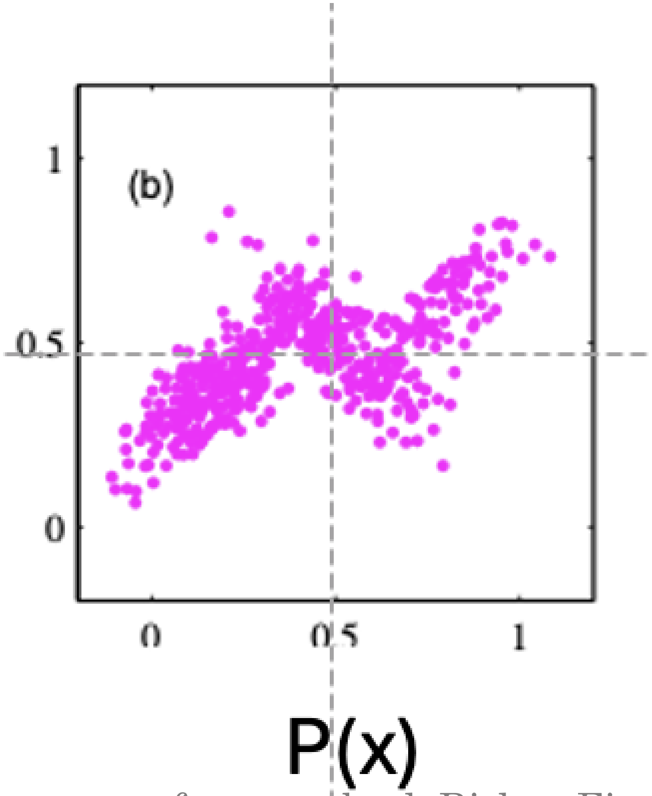


from textbook Bishop Fig. 9.5

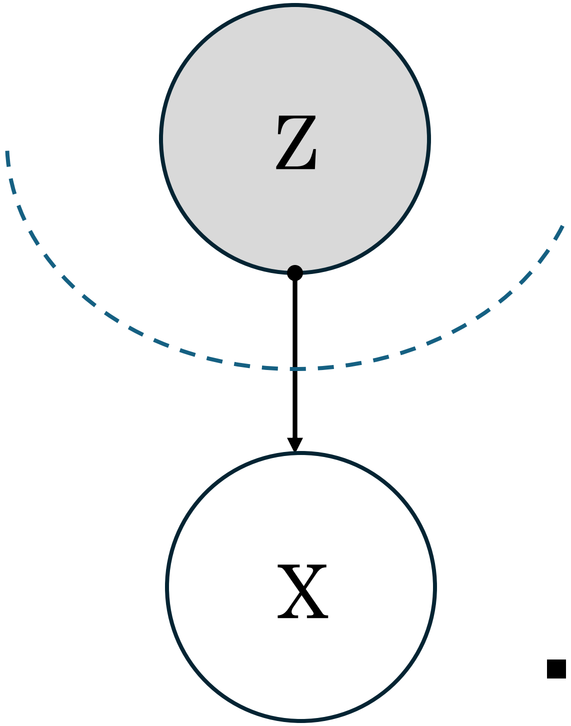
- density modeling $P(X)$
  - what R.Vs
  - how they are connected
  - CPTs

[2] GMM (motivation)

Q: what if we introduce a new R.V but invisible to us?



from textbook Bishop Fig. 9.5

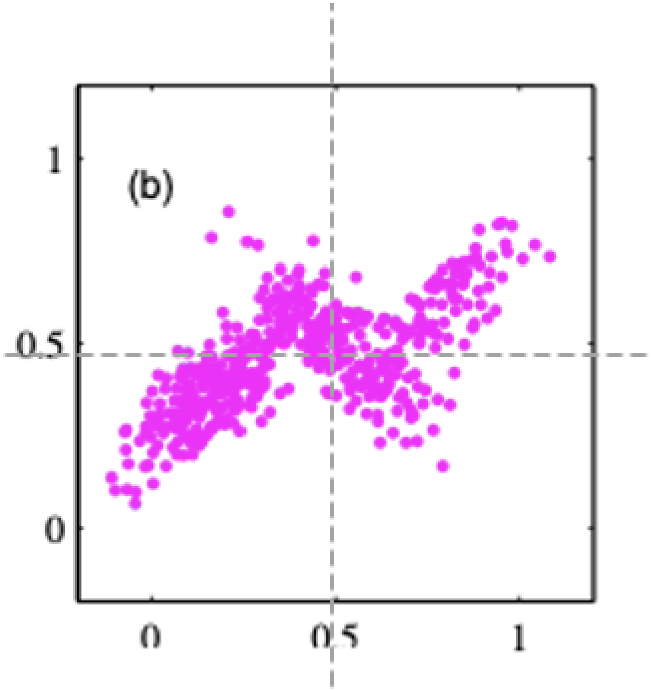


[with Z]

- density modeling $P(X)$
  - what R.Vs
  - how they are connected
  - CPTs

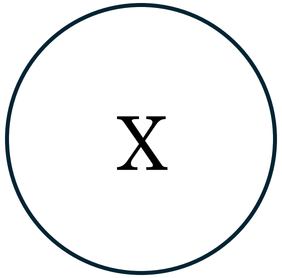
[3] GMM (hierarchical structure with a latent variable)

which model is structured so <u>efficient</u> and <u>scalable</u>?
Is the second model, Z accessible?



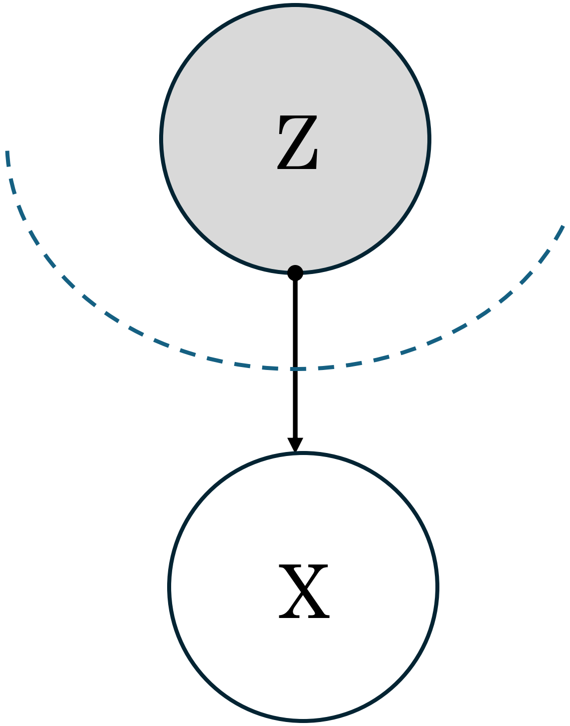
P(x)

from textbook Bishop Fig. 9.5



[without Z]

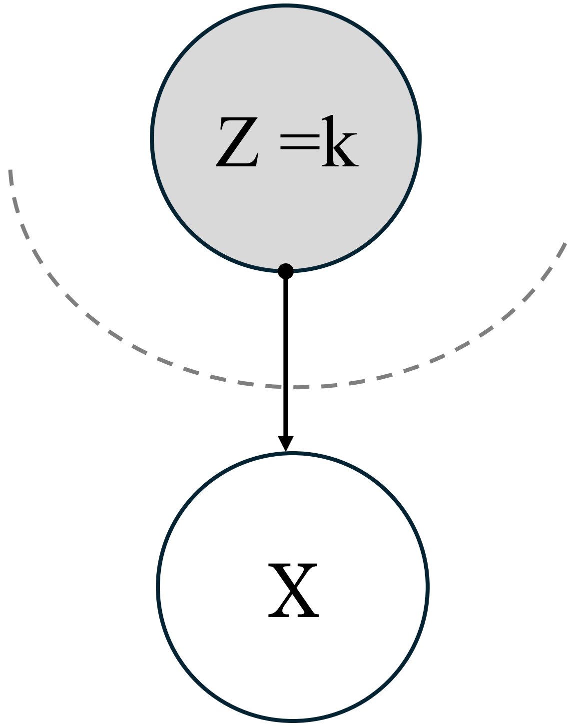
vs.



[with Z]

[4] GMM (parameters)

- Bayesian Network Representation of GMM

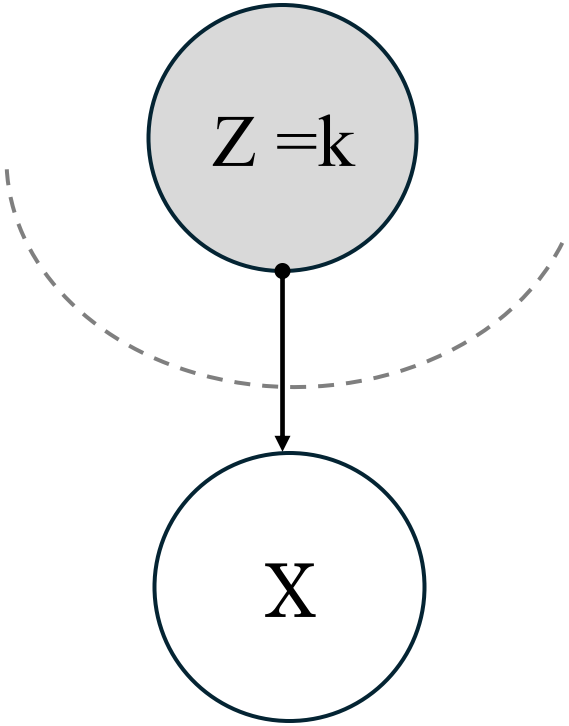


- $P(Z = k) = \pi_k$
- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

Q: $P[X]$?

[5] GMM (parameters)

- Bayesian Network Representation of GMM



- $P(Z = k) = \pi_k$
- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

$$P(X) = \sum_{Z=k} P(X, Z) = \sum_{Z=k} P(Z = k) P(X|Z = k)$$
$$= \sum_{Z=k} \pi_k \mathcal{N}(\mu_k, \Sigma_k)$$

Applications of Gaussian **M**ixture **M**odel (GMM)

- new data simulation
- data compression

[1] GMM (Data Generation)

Q: how can we generate Gaussian Mixture (GMM) samples?

[2] GMM Q: how can we generate Gaussian Mixture (GMM) samples?

[Generate N GMM samples]

- suppose Z ~ K-ary discrete R.V
- $P(X|Z=k)$ ~ Gaussian R.V ~ $N(\mu_k, \Sigma_k)$

[3] GMM Q: how can we generate Gaussian Mixture (GMM) samples?

[Generate N GMM samples]

- suppose Z ~ K-ary discrete R.V
- $\mathrm{P}(X|Z=k)$ ~ Gaussian R.V ~ $N(\mu_k, \Sigma_k)$

(1) generate the samples for $Z \in \{0, 1, 2\}$.

(2) depending on z,
choose a Gaussian density and generate a sample x according to the Gaussian

(3) repeat (1) and (2) for N times

[4] GMM (generation example)

(1)generate the samples for $Z \in \{0, 1, 2\}$.

$$\boldsymbol{P(z=0):\ 0.3, P(z=1):\ 0.2, P(z=2):\ 0.5}$$

(2) depending on z, choose the a Gaussian density.

$$\boldsymbol{G_0: \left(\begin{bmatrix} 0 \\ 4 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}\right)}$$

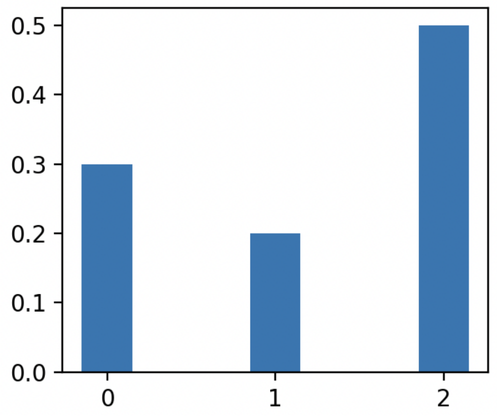
$$\boldsymbol{G_1: \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}\right)}$$

$$\boldsymbol{G_2: \left(\begin{bmatrix} 0 \\ -4 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}\right)}$$

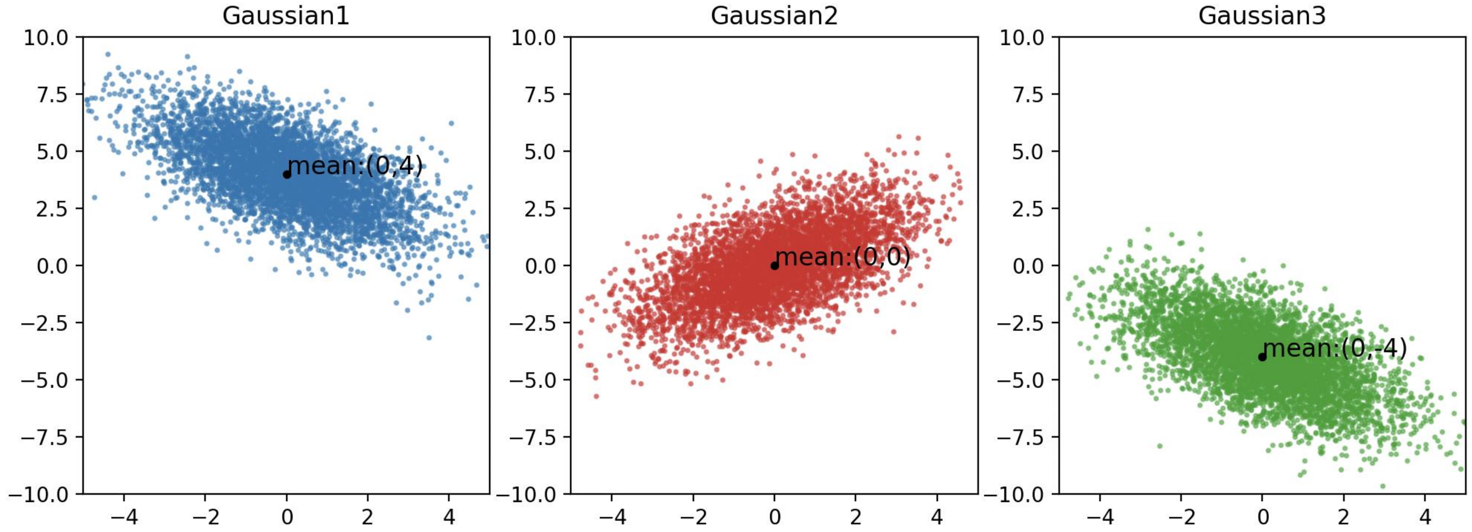
[5] GMM (generation example)

How can we generate Gaussian Mixture Modeling (GMM) samples?

(1) generate the samples for $Z \in \{0, 1, 2\}$.



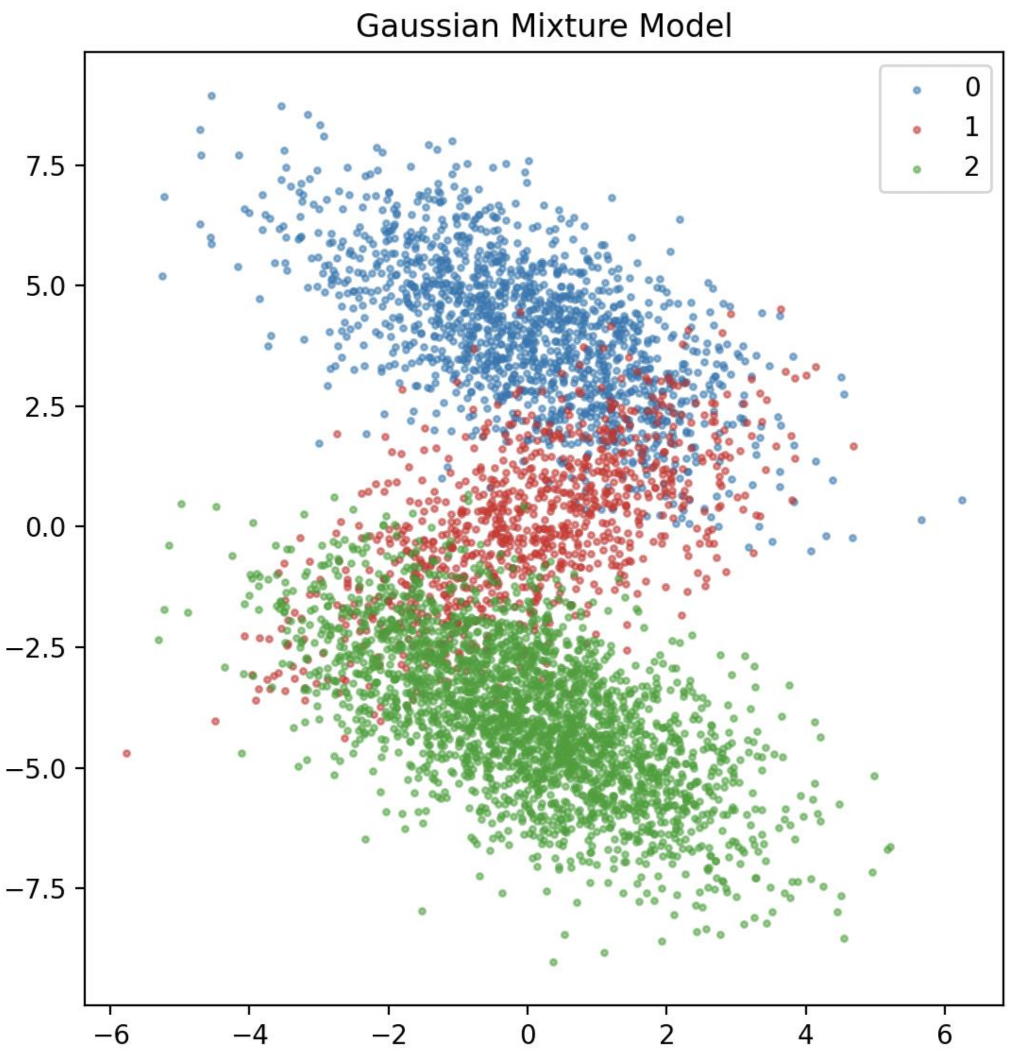
(2) depending on z, choose the a Gaussian density.



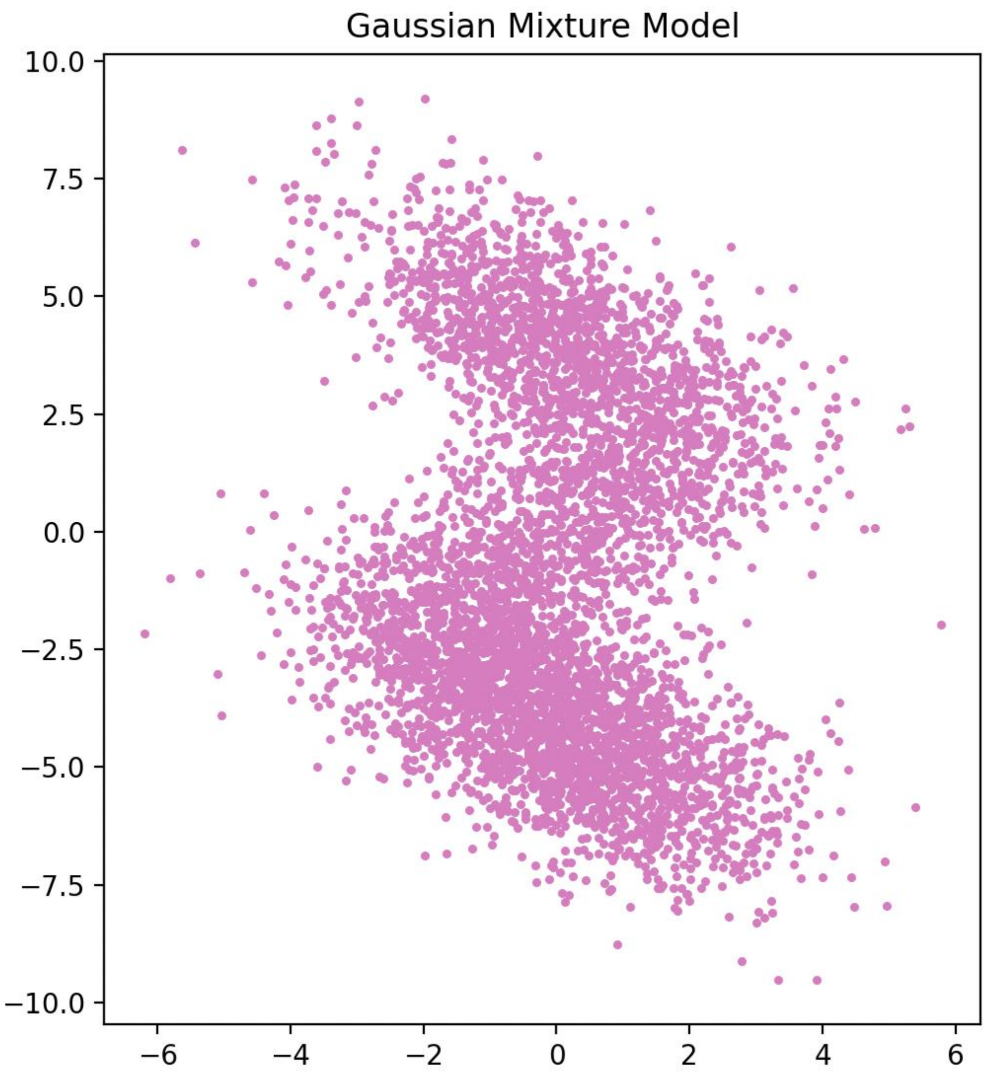
[6] GMM (generation example)

Q: Can you imagine the collective GMM samples (10,000)?

[7] GMM (example)

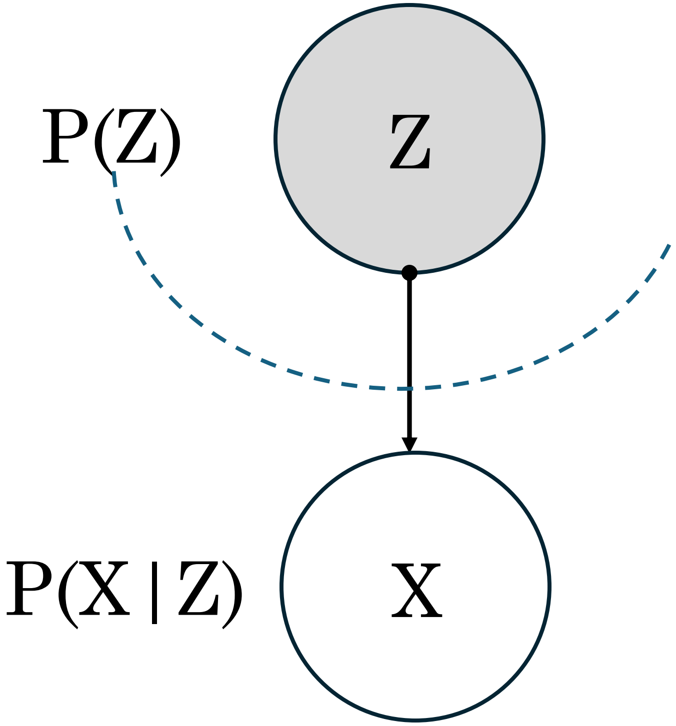


[when z is known]



[when z is unknown]

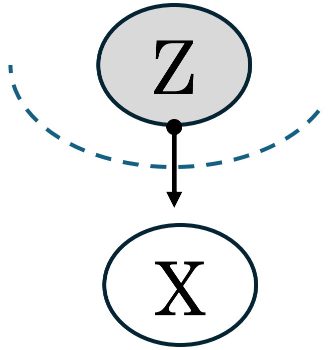
[8] GMM (data compression/representation)



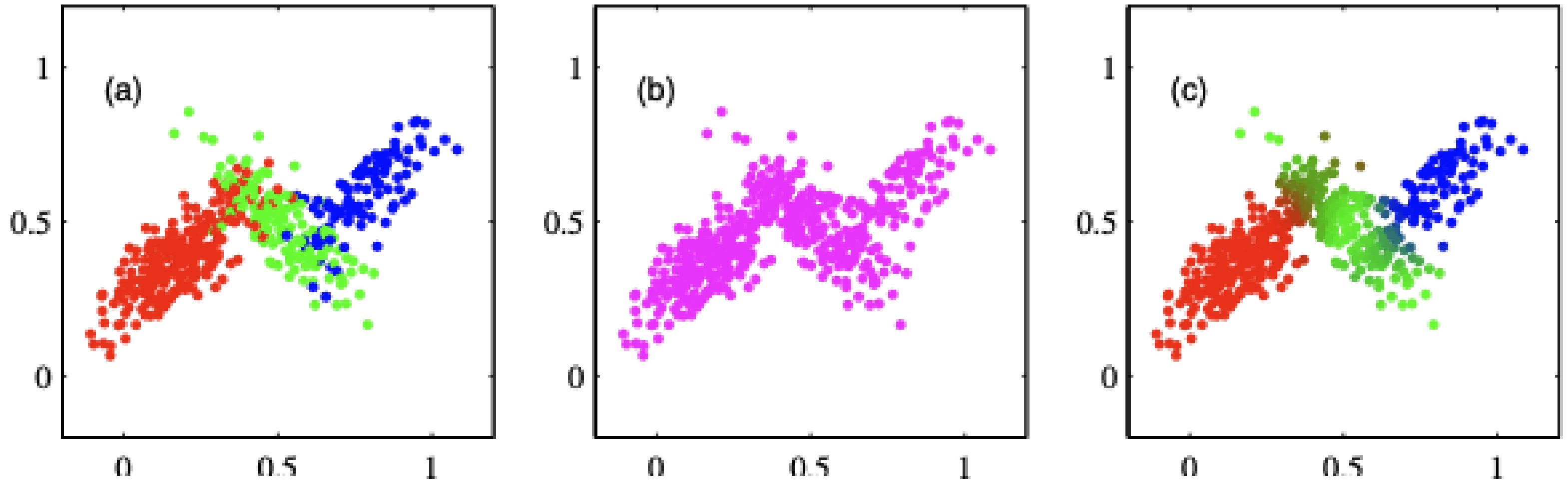
once we set this model,
we could compute the followings.

- $P(X)$
- $P(Z = k \,|x)$
  * valuable information
  * prediction about invisible factors.
  * useful for clustering (when z is discrete)

[9] GMM (data compression/representation)



$$P(Z, X) = P(Z) \cdot P(X|Z)$$



$P(Z|x)$

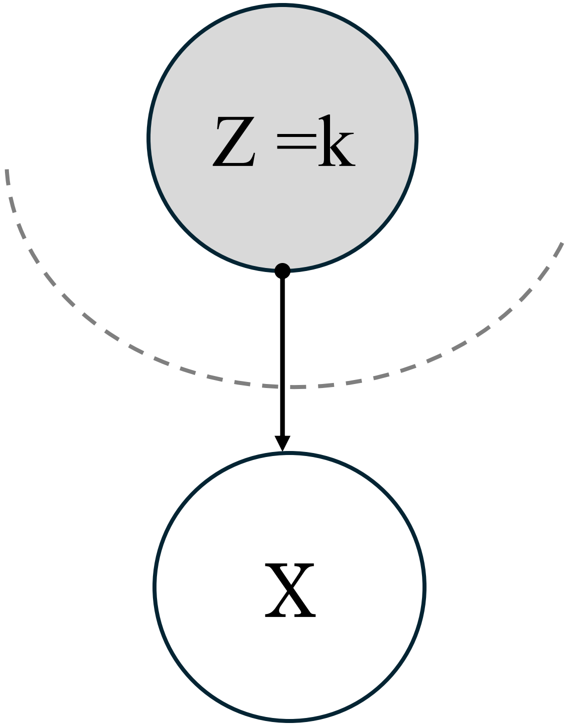
- Representation will be $Z = \text{argmax}\ P(Z|x)$
- the representation will be one of colors: red / green / blue.

from textbook Bishop Fig. 9.5

- EM for GMM: “Given a data set”, how can we learn the parameters?

** GMM (parameters)

- Bayesian Network Representation of GMM



- $P(Z = k) = \pi_k$
- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

$$
\begin{aligned}
P(X) &= \sum_{Z=k} P(X, Z) = \sum_{Z=k} P(Z = k) P(X|Z = k) \\
&= \sum_{Z=k} \pi_k \mathcal{N}(\mu_k, \Sigma_k)
\end{aligned}
$$

** GMM

Q: Learning a GMM:

is to learn the parameters : π, μ, Σ (as structure is given)

but we cannot access Z (imaginary / invisible).

how can we learn the parameters?

what method can we use?

[1] EM for GMM (likelihood & lower bound)

$$\log P(D|\theta) = \sum_{n=1}^{N} \log \sum_{Z_n=k} P(Z_n, X_n|\theta)$$

$$= \sum_{n=1}^{N} \log \sum_{Z_n=k} \frac{P(Z_n, X_n|\theta) q(Z_n = k)}{q(Z_n = k)}$$

$$= \sum_{n=1}^{N} \log E[\frac{P(Z_n, X_n|\theta)}{q(Z_n = k)}]$$

- expectation over $\mathrm{q}(Z_n = k)$

$$\geq \sum_{n=1}^{N} E[\log (\frac{P(Z_n, X_n|\theta)}{q(Z_n = k)})]$$

$$= \sum_{n=1}^{N} \sum_{k} q(Z_n = k) \cdot (\log \pi_k + \log \mathcal{N}(x_n|\mu_k, \Sigma_k)) + \sum_{n=1}^{N} H(q(Z_n = k))$$

Q: what $q(z_n = k)$ gives a tight bound?

[2] EM for GMM (Expectation E step)

parameters at time t

$$
\begin{aligned}
q(Z_n = k) &= P(Z_n = k | X_n = x_n, \pi_k(t), \mu_k(t), \Sigma_k(t)) \\
&= \frac{P(Z_n = k, X_n = x_n | \pi_k(t), \mu_k(t), \Sigma_k(t))}{P(X_n = x_n | \pi_t, \mu_t, \Sigma_t)} \\
&= \frac{P(Z_n = k | \pi_k(t), \mu_k(t), \Sigma_k(t)) P(X_n = x_n | Z_n = k, \pi_k(t), \mu_k(t), \Sigma_k(t))}{P(X_n = x_n | \pi_k(t), \mu_k(t), \Sigma_k(t))} \\
\boxed{\gamma_{nk}(\pi_k(t), \mu_k(t), \Sigma_k(t))} &= \frac{P(Z_n = k | \pi_t, \mu_t, \Sigma_t) P(X_n = x_n | Z_n = k, \pi_k(t), \mu_k(t), \Sigma_k(t))}{P(X_n = x_n | \pi_k(t), \mu_k(t), \Sigma_k(t))}
\end{aligned}
$$

- **responsibility** that cluster K takes for data point n

- M step
- [1] update μ_k

$$\boxed{\sum_{n=1}^{N}\sum_{k} q(Z_n = k)\cdot(\log \pi_k + \log \mathcal{N}(x_n|\mu_k,\Sigma_k)) + \sum_{n=1}^{N} H(q(Z_n = k))}$$

+ this is the lower bound we computed in the previous slide.

$$L = \sum_{n=1}^{N}\sum_{k} q(Z_n = k)\cdot(\log \pi_k + \log \mathcal{N}(x_i|\mu_k,\Sigma_k)) + \sum_{n=1}^{N} H(q(Z_n = k))$$

$$L = \sum_{n=1}^{N}\sum_{k} r_{nk}\cdot(\log \pi_k + \log \mathcal{N}(x_n|\mu_k,\Sigma_k)) + \sum_{n=1}^{N} H(r_{nk})$$

$$\frac{\partial L}{\partial \mu_k} = \frac{\partial}{\partial \mu_k}\sum_{n=1}^{N}(-1/2\gamma_{nk}(x_n-\mu_k)^t\Sigma^{-1}(x_n-\mu_k))$$

$$\frac{\partial L}{\partial \mu_k} = \sum_{n=1}^{N}\gamma_{nk}\Sigma^{-1}(x_n-\mu_k) = 0$$

$$\boxed{\mu_k = \frac{\sum_{n=1}^{N}\gamma_{nk}x_n}{\sum_{n=1}^{N}\gamma_{nk}}}$$

$\mathbf{y}$	$\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$
$\mathbf{Ax}$	$\mathbf{A}^T$
$\mathbf{x}^T\mathbf{A}$	$\mathbf{A}$
$\mathbf{x}^T\mathbf{x}$	$2\mathbf{x}$
$\mathbf{x}^T\mathbf{Ax}$	$\mathbf{Ax}+\mathbf{A}^T\mathbf{x}$

- M step of GMM
- [2] update Σ_k

$$L = \sum_{n=1}^{N} \sum_{k} r_{nk} \cdot (\log \pi_k + \log \mathcal{N}(x_n|\mu_k, \Sigma_k)) + \sum_{n=1}^{N} H(r_{nk})$$

$$\frac{\partial L}{\partial \Sigma_k^{-1}} = \frac{\partial}{\partial \Sigma_k^{-1}} \sum_{n=1}^{N} (-1/2(\gamma_{nk}(x_n - \mu_k)^t \Sigma^{-1} (x_n - \mu_k) - \log \sqrt{(2\pi)^n \cdot 1/|\Sigma^{-1}|}))$$

$$= \sum_{n=1}^{N} -1/2 \gamma_{nk}(x_n - \mu_k)^t (x_n - \mu_k) + \gamma_{nk} 1/2 \cdot \Sigma_k = 0$$

$$\boxed{\Sigma_k = \frac{\sum_{n=1}^{N} \gamma_{nk}(x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^{N} \gamma_{nk}}}$$

$$\frac{\partial}{\partial A} \log |A| = A^{-T}$$

$$\frac{\partial}{\partial A} x^T A x = \frac{\partial}{\partial A} \mathrm{tr}[x x^T A] = [x x^T]^T = x x^T$$

- M step of GMM
  [3] update π_k

$$L = \sum_{n=1}^{N} \sum_{k} r_{nk} \cdot (\log \pi_k + \log \mathcal{N}(x_n|\mu_k, \Sigma_k)) + \sum_{n=1}^{N} H(r_{nk}) + \lambda^* (\sum_k \pi_k - 1)$$

$$\frac{\partial L}{\partial \pi_k} = \frac{\partial}{\partial \pi_k} \sum_{n=1}^{N} (\gamma_{nk} \log \pi_k) + \lambda^* \pi_k$$

$$\frac{\partial L}{\partial \pi_k} = \sum_{n=1}^{N} \gamma_{nk}/\pi_k + \lambda^* = 0$$

$$\pi_k^* = \frac{\sum_n^N \gamma_{nk}}{\lambda^*}$$

$$\boxed{\pi_k^* = \frac{\sum_n^N \gamma_{nk}}{N}}$$

- *we know* $\sum_k \gamma_{nk} = 1$

$$\frac{\partial L}{\partial \lambda} = \sum_k \pi_k = 1$$

$$\sum_k \pi_k = \sum_k \frac{\sum_n^N \gamma_{nk}}{\lambda^*} = 1$$

$$N/\lambda^* = 1$$

$$\lambda^* = N$$

[3] EM for GMM (Maximization M step)

- $\mu_k = \dfrac{\sum_{n=1}^{N} \gamma_{nk} x_n}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\Sigma_k = \dfrac{\sum_{n=1}^{N} \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\pi_k = \dfrac{\sum_{n}^{N} \gamma_{nk}}{N}$

- M steps for GMM

they are the sample means
weighted by responsibility
$\gamma_{nk} = P[\, Z_k = \mathrm{k} \,|X_k = x_k, \mu_k(t), \pi_k(t), \Sigma_k\,(t)]$!!

[4] EM for GMM (summary)

[1] **E step:** compute the tight lower bound at current parameters

compute $\gamma_{nk} = P[\, Z_n = k \mid X_n = x_k, \mu_k(t), \pi_k(t), \Sigma_k\,(t)]$

[2] **M step:** update parameters using the responsibility

- $\mu_k = \dfrac{\sum_{n=1}^{N} \gamma_{nk} x_n}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\Sigma_k = \dfrac{\sum_{n=1}^{N} \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\pi_k = \dfrac{\sum_{n}^{N} \gamma_{nk}}{N}$

** retake the previous example (sprinkler, rain, cloud)

Let’ s go back to the problem of
Learning the parameters of Bayesian network.

** retake the previous example (sprinkler, rain, cloud)

what if data is partially observed?
Suppose we forgot to collect cloud information.
Can we still estimate *p or q* : P[rain| cloud +] or P[rain| cloud -]?

** retake the previous example (sprinkler, rain, cloud)

- E step : compute $\gamma_{nk}(t) = P[Cloud\ (n) = k \mid Rain\ (n)\ \ and\ S\ (n)\ ]$

- M step: update the parameters

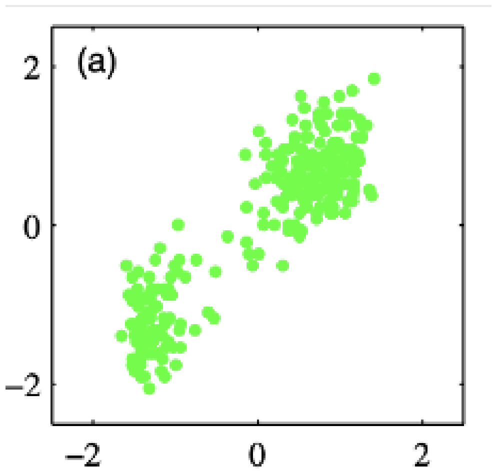
- $P(C)(t+1) = \dfrac{\sum_{n=1}^{N} \gamma_{nk}(t)}{N}$

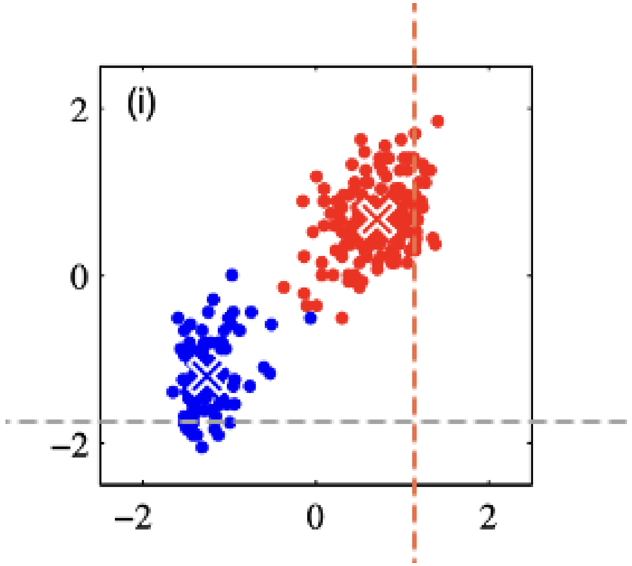
- $P(S:+, R:+|C:+)(t+1) = \dfrac{\sum_{n=1}^{N} \gamma_{nk}\delta(x_n = S:+, R:+)}{\sum_{n=1}^{N} \gamma_{nk}}$

- K means Clustering (partition the data set into K of clusters as $\#K$ given)

[1] K-means Clustering (goal)

[Clustering]:
the goal is to learn a finite a prototype μ_k, the center of the clusters so enable to assign each data points x_n to one of the clusters





- the K-means algorithm can be understood in the context of GMM and EM, but approximation & simplification.

[2] K-means Clustering (EM for GMM)

[1] E step

compute $\gamma_{nk} = P[\, Z_k = \mathrm{k} \,|X_k = x_k, \, \mu_k(t), \pi_k(t), \Sigma_k\,(t)]$

[2] M step

- $\mu_k = \frac{\sum_{n=1}^{N} \gamma_{nk} x_n}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\Sigma_k = \frac{\sum_{n=1}^{N} \gamma_{nk} (x_n - \mu_k)^t (x_n - \mu_k)}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\pi_k = \frac{\sum_{n}^{N} \gamma_{nk}}{N}$

[3] K-means Clustering

[1] E step

$$= \frac{P(Z_n = k|\pi_k(t), \mu_k(t), \Sigma_k(t))\boxed{P(X_n = x_n|Z_n = k, \pi_k(t), \mu_k(t), \Sigma_k(t))}}{P(X_n = x_n|\pi_k(t), \mu_k(t), \Sigma_k(t))}$$

compute $\gamma_{nk} = 1 \;\; when\ k = argmin\_\{k\}\ P[\ Z_k = \mathrm{k}\ |X_k = x_k,\ \mu_k(t), \pi_k(t), \Sigma_k\ (t)]$

$\gamma_{nk} = 0\ o.w$

prob $\propto ||x_n - \mu_k||$ ** 2

[2] M step

- $\mu_k = \frac{\sum_{n=1}^{N} \gamma_{nk} x_n}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\Sigma_k = \frac{\sum_{n=1}^{N} \gamma_{nk}(x_n - \mu_k)^t(x_n - \mu_k)}{\sum_{n=1}^{N} \gamma_{nk}}$

- $\pi_k = \frac{\sum_{n}^{N} \gamma_{nk}}{N}$

GPT comparison☺

Aspect	K-means	EM for GMM
Assignment	Hard	Soft (probabilistic)
Distribution Assumption	Implicit: spherical Gaussians with equal variance	Explicit: full Gaussian (can have different covariances)
Cluster shape	Spherical	Elliptical
Parameters	Only means (centroids)	Means, covariances, and weights

Q: do you think K-mean will show the monotonic increasing behavior?

[5] K-means Clustering (algorithm, #k given / assumed)

(1) choose some initial values for μ_k

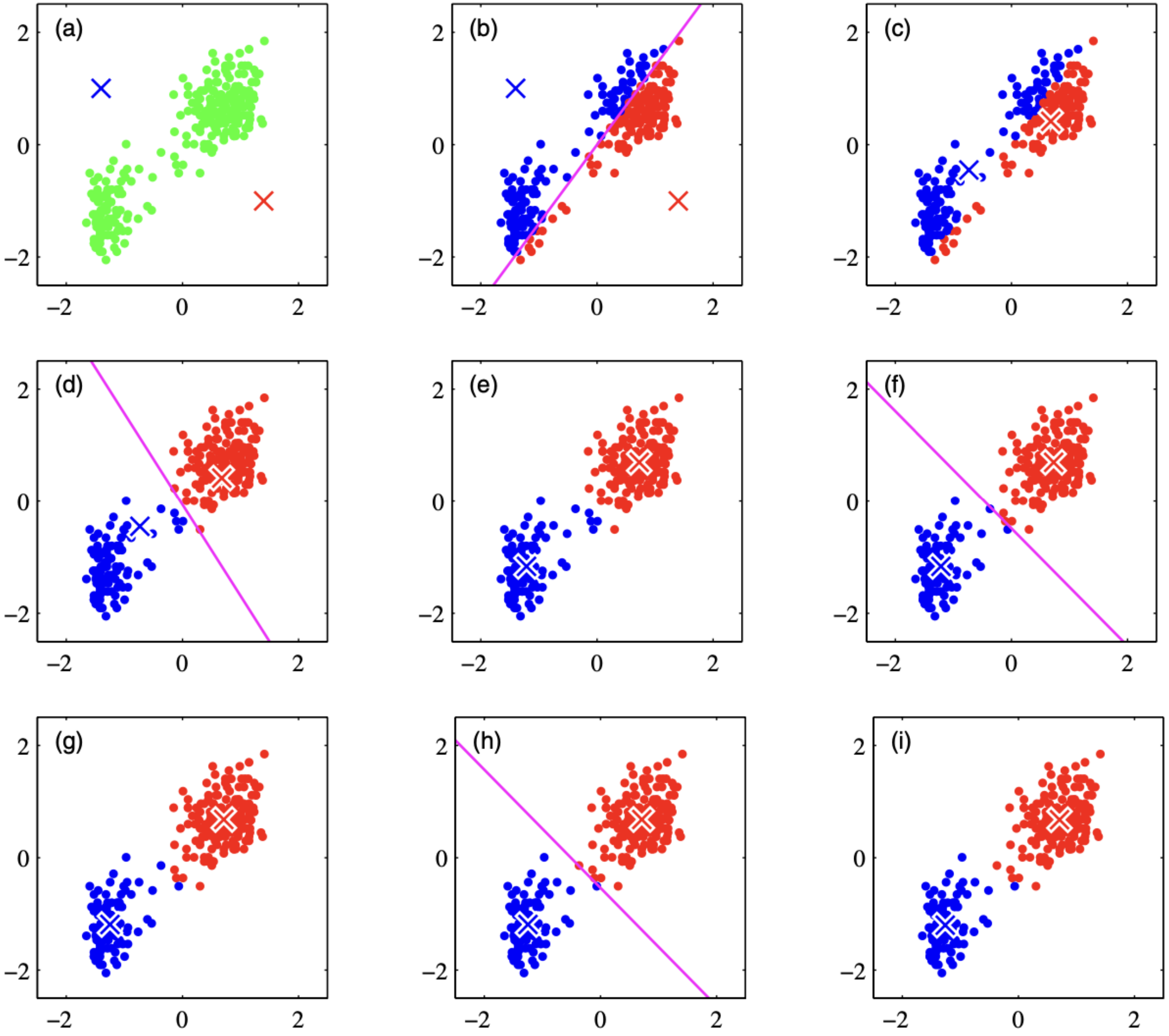
(2) E step

- $\gamma_{nk} = 1 \ when\ k = argmin\_\{k\}||x_n - \mu_k(t)|| ** 2$
- $\gamma_{nk} = 0\ o.w$

(3) M step

- $\mu_k = \frac{\sum_{n=1}^{N} \gamma_{nk} x_n}{\sum_{n=1}^{N} \gamma_{nk}}$

[6] K-means Clustering (illustration of K-means progression)

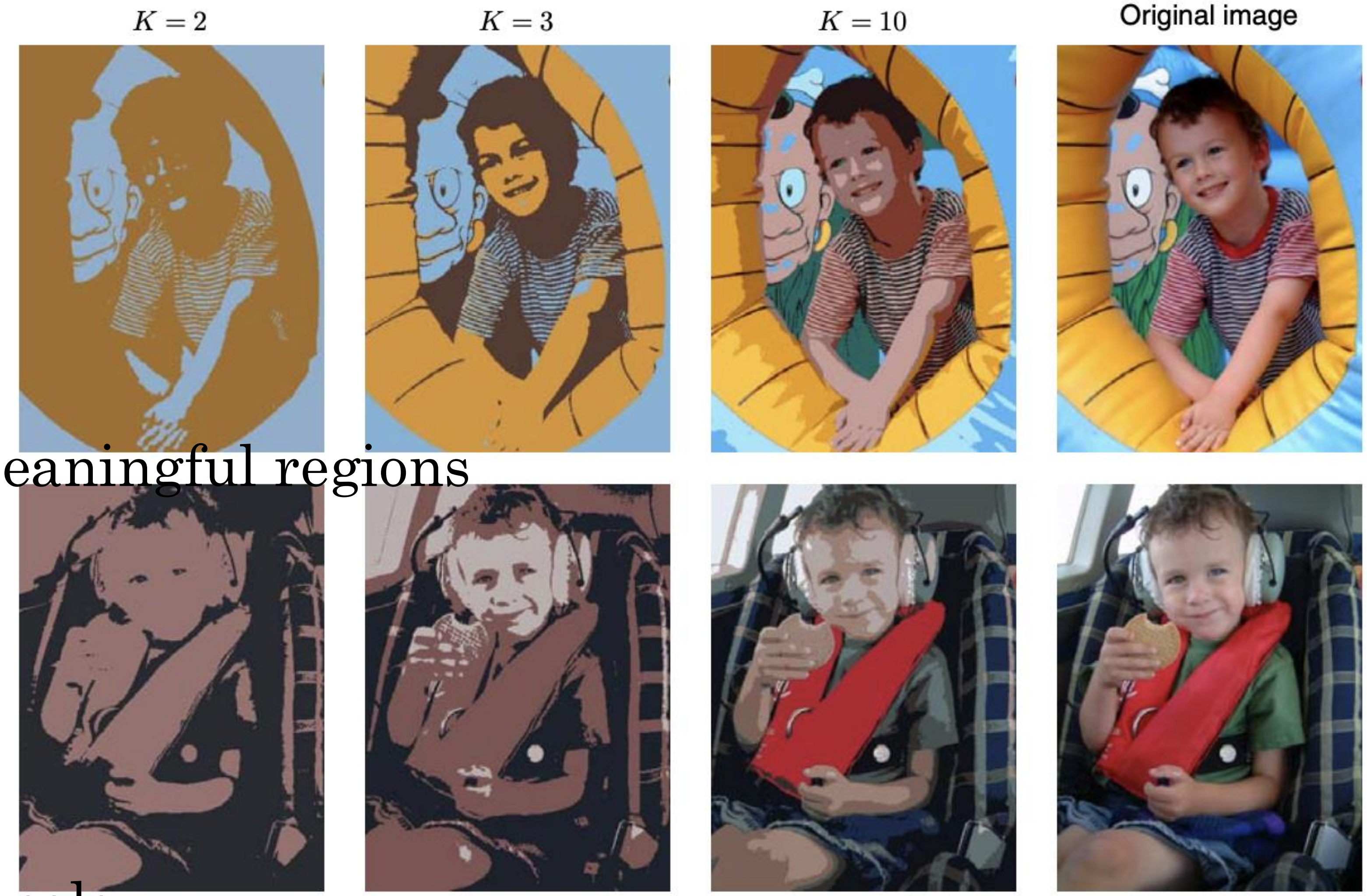


**continue until there is no further changes in the assignment.

from textbook Bishop Fig. 9.1

[7] K-means Clustering (K-means examples)

- image segmentation
- image compression



**labeling
partitions the image into meaningful regions
not sophisticated one, why?

**compression
images with fewer distinct colors